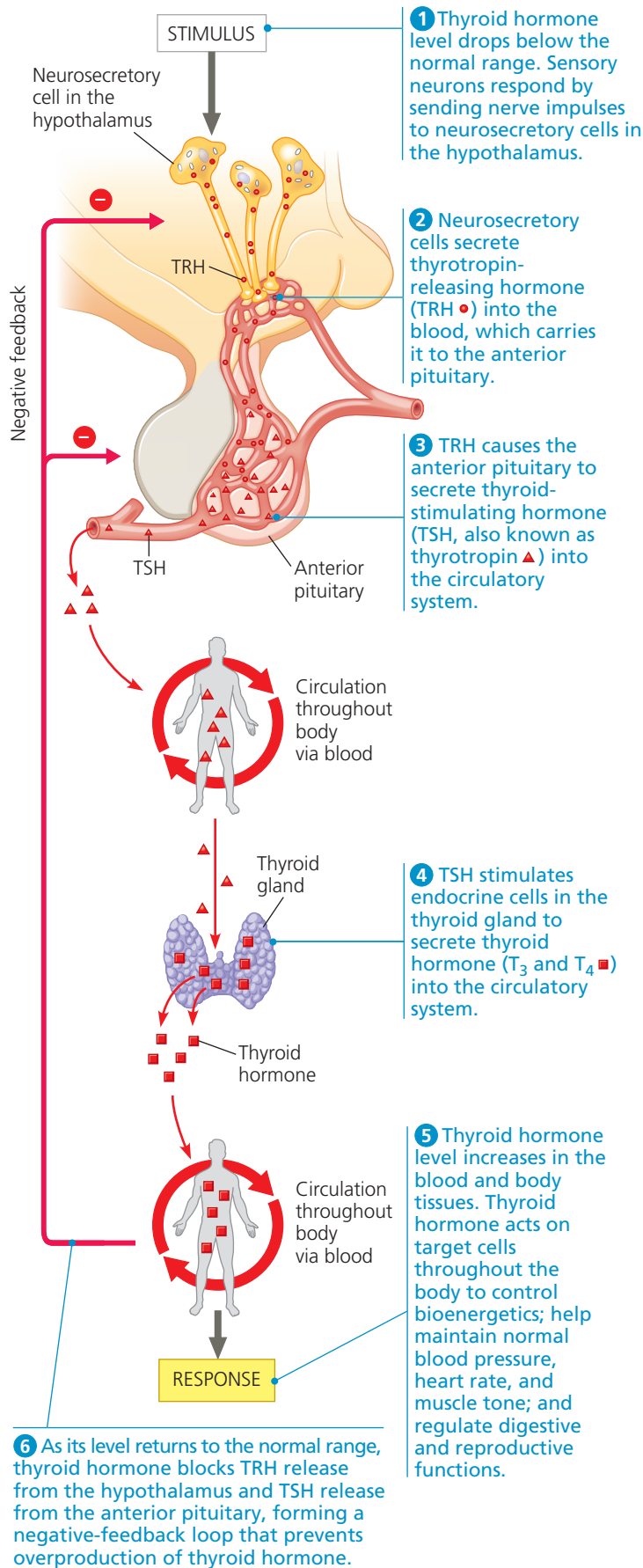


▼ **Figure 45.16 Regulation of thyroid hormone secretion: a hormone cascade pathway.**



pathway that regulates thyroid hormone release. If the level of thyroid hormone in the blood drops, the hypothalamus secretes thyrotropin-releasing hormone (TRH), causing the anterior pituitary to secrete thyrotropin, a tropic hormone also known as thyroid-stimulating hormone (TSH). TSH in turn stimulates the **thyroid gland**, an organ in the neck consisting of two lobes on the ventral surface of the trachea. The thyroid gland responds by secreting thyroid hormone, which increases metabolic rate.

As with other hormone cascade pathways, feedback regulation often occurs at multiple levels. For example, thyroid hormone exerts negative feedback on the hypothalamus and on the anterior pituitary, in each case blocking release of the hormone that promotes its production (see Figure 45.16).

Disorders of Thyroid Function and Regulation

Disruption of thyroid hormone production and regulation can result in serious disorders. One such disorder reflects the unusual chemical makeup of thyroid hormone, the only iodine-containing molecule synthesized in the body. *Thyroid hormone* is actually a pair of very similar molecules derived from the amino acid tyrosine. *Triiodothyronine* (T_3) contains three iodine atoms, whereas *tetraiodothyronine*, or *thyroxine* (T_4), contains four (see Figure 45.4).

Although iodine is readily obtained from seafood or iodized salt, people in many parts of the world lack enough iodine in their diet to synthesize adequate amounts of thyroid hormone. With only a low blood level of thyroid hormone, the pituitary receives no negative feedback and continues to secrete TSH. An elevated TSH level in turn causes the thyroid gland to enlarge, resulting in **goiter**, a marked swelling of the neck.

Hormonal Regulation of Growth

Growth hormone (GH), which is secreted by the anterior pituitary, stimulates growth through both tropic and non-tropic effects. A major target, the liver, responds to GH by releasing *insulin-like growth factors* (IGFs), which circulate in the blood and directly stimulate bone and cartilage growth. (IGFs also appear to play a key role in aging in many animal species.) In the absence of GH, the skeleton of an immature animal stops growing. GH also exerts diverse metabolic effects that tend to raise the blood glucose level, thus opposing the effects of insulin.

Abnormal production of GH in humans can result in several disorders, depending on when the problem occurs and whether it involves hypersecretion (too much) or hyposecretion (too little). Hypersecretion of GH during childhood can lead to **gigantism**, in which the person grows unusually tall but retains relatively normal body